卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-GL-0105 30-B003701S-Z0105

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

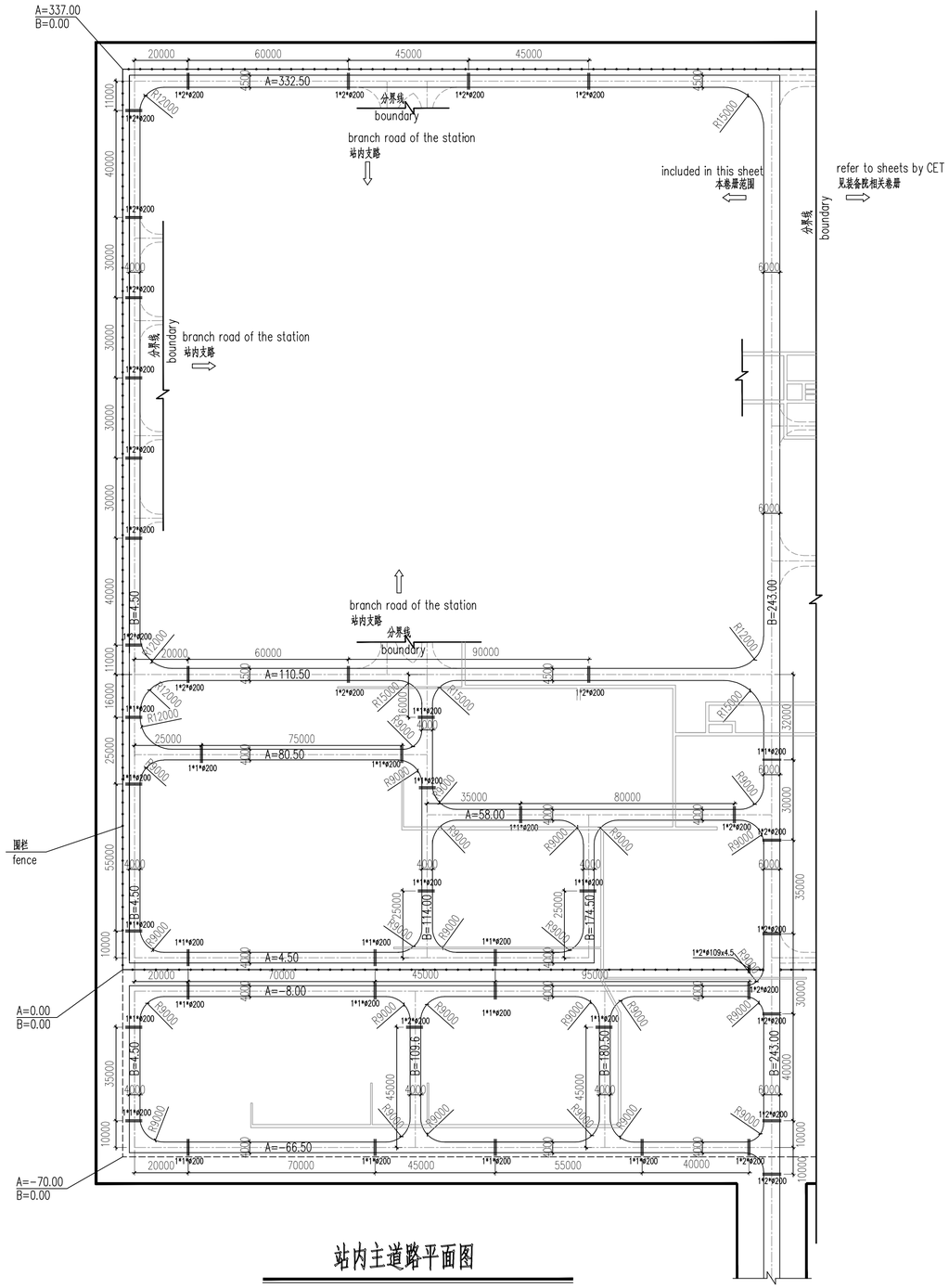
总图 专业 第 01 卷 第 05 册
General Layout SPECIALITY VOLUME BOOK

卷册名称 站内主道路
Volume name Main road of the station

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	孙晨曦
图纸 2 张 DRAWING 2 NOS.	打印件 0 本 PRINTED COPIES 0 NOS.

序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-Z0105-01	站内主道路及埋管平面图 layout of main road of the station and buried pipe	A	1		
2	30-B003701S-Z0105-02	站内主道路详图 detail of main road of the station	A	1		

注:



站内主道路平面图

layout of main road of the station

N

说明
Notes:

1.本图采用巴基斯坦UTM-43N平面坐标系，EGM2008高程系统。
1.Coordinate system: Pakistan UTM-43N. Elevation system: EGM2008

2.相对平面坐标以米(m)为单位，尺寸以毫米(mm)为单位。
2.Relative coordinates system is measured in meter.
Dimensions are measured in millimeters.

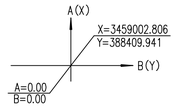
3.场地平均设计标高为193.40m。
3.Site level is equal to 193.40m.

4.站内道路为郊区型道路，道路中心标高193.60m。
4.Interior Road is suburban type, the center level of interior road is 193.60m

5.站内道路设计等级为四级，设计时速15km/h,设计荷载以单轴100kN为设计荷载，设计年限10年。
5.Designed level: 4 Designed speed: 15km/h;Designed single axle load: 100kN;Designed period : about 10 years.

6.未注明道路的转弯半径均为9.0米。
6.The radius of interior roads which are not indicated is 9.0m.

7.本图建筑(A、B)坐标与UTM-43N平面坐标系关系如下所示:
7.The relationship between relative coordinate system
and UTM-43N system is as below:



8.本工程初步设计尚未审定,未通过NIDC审核，最终方案还存在修改的可能性，
施工前请联系业主与设计确认，由此引起的方案变更设计不予负责。
8.Because of preliminary design documents are not approved by NIDC,
the project solution has prossibility to be modified.
The design contractor has no responsibility for the project solution changed,
if the construction contractor not get authorized by project owner and
design contractor before construction started.

9.电缆沟过道路做法见Z0109册图。
9.For the details of cable trench crossing the road, refer to drawings Z0109.

10.过道路电缆沟及埋管等全部施工完成后，方可浇筑道路面层。
10.It is only after the construction of cable trenches crossing the road and buried pipes
that the construction of roads are allowed to start.

LEGEND(图例)

A=-70.00	Location coordinate 定位坐标
	Fence 围栏
	Main road 主道路
	Buried pipes 过道路埋管
	Road (refer to other sheet) 道路(见相关卷册)
	Cable trench 电缆沟

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - B C I - D D - G L - 0 1 0 5 - 0 0 1 A

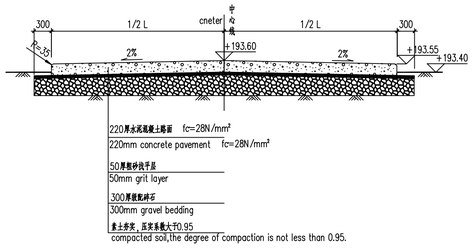
电气		
专业 Speciality	会签 Sign	日期 Date

	姓名 Name	日期 Date
设计 Designed	杨晨乘	2017.10.30
校核 Checked	周七璐	
审核 Reviewed	吴海强 毛东方	
批准 Approved	杨涛	
比例 Scale	1:1000	图号 DWG NO. 30-B003701S-Z0105-001

站内主道路及埋管平面图

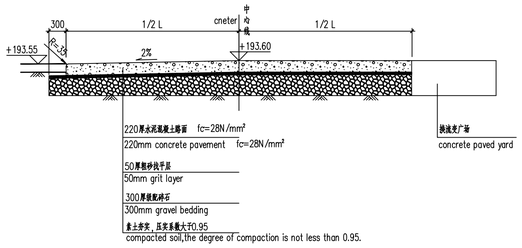
layout of main road of the station
and buried pipe

版本 REV. A



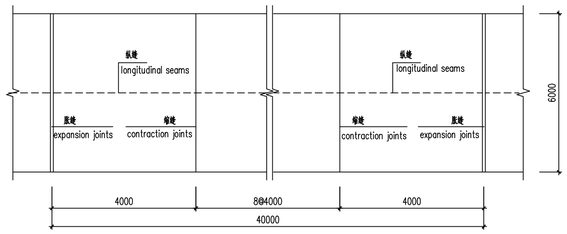
站内道路横断面图1

section of road No.1



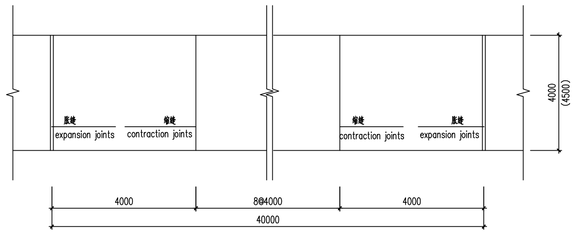
站内道路横断面图2(适用与换流变广场连接处)

section of road No.2 (for joint with concrete paved yard)



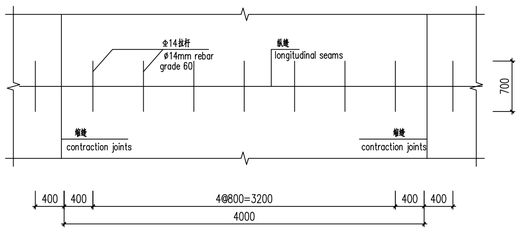
典型伸缩缝布置图1

typical arrangement of expansion joints No.1



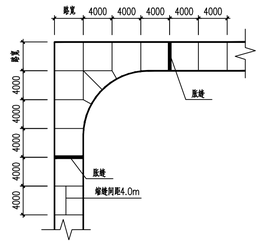
典型伸缩缝布置图2

typical arrangement of expansion joints No.2



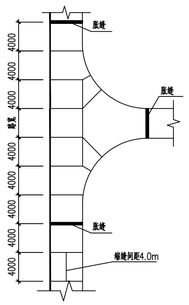
纵缝平面布置

typical arrangement of longitudinal seams



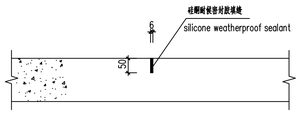
"L"型路口分缝图

"L" style cross road joint detail



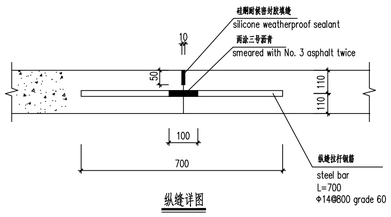
"T"型路口分缝图

"T" style cross road joint detail



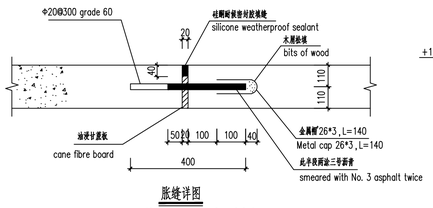
缩缝详图

section of contraction joints



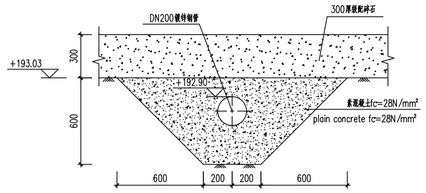
纵缝详图

section of longitudinal seams



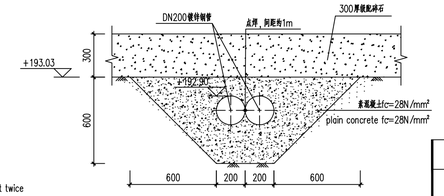
胀缝详图

section of expansion joints



1Ø200过道路理管

1Ø200 buried pipe



2Ø200过道路理管

2Ø200 buried pipe

说明：

Note：

1.碎石基层：颗粒最大尺寸不宜超过60mm，石料中不得含有有机质。

1.Gravel bedding:the maximum size should not exceed 60mm, and does not contain organic material and shall not be greater than 0.8 times the thickness of the layer.No organic matter in gravel.

2.基层压实要求：应在最佳含水量时进行碾压。基层检验合格后方可进行面层的施工。

2.Compaction should be carried out in the best water content of the roller. The construction of the surface layer can be carried out after the inspection.

3.站内道路表层耕植土应去除，路基土须用重型压路机分层碾压密实，分层厚度不大于30cm，压实度不小于0.95。

3.The surface of the road surface should be removed, the subgrade soil must be compacted with heavy roller, the thickness of every layer is not more than 30cm, the degree of compaction is not less than 0.95.

4.在基层碾压此壓后,需检查基层的强度和质量,基层强度以基层顶面的当量回弹模量值为强度检查指标(采载板法),其当量回弹模量值不得小于80MPa。

4.After the roller compaction, the strength and quality of the base should be checked, the value of resilient modulus of the base surface should exceed 80MPa.

5.道路分缝以面板不出现锐角为原则，现场可根据实际情况适当调整

5.The arrangement of deformation joints of concrte slabs shall be based on principle of no acute angle and the joints can be adjusted according to the actual situation.

6.埋管为镀锌钢质，标注含义如下,规格为209x4.5

1*2*φ200
- 埋管内径(mm)
- 每层的埋管数数
- 埋管层数

6.The material of buried pipes are galvanized steel and the symbol indicates as follows, the size is 209x4.5

1*2*φ200
- inner diameter(mm)
- number of pipes for every layer
- number of layers

7.过道路埋管顶标高+192.90米，埋管两端伸出道路基础500mm。

7.Elevation of the top of buried pipes is +192.90m. And for each end of the pipes, make sure it stretches out for 500mm.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

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±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - B C I - D D - G L - 0 1 0 5 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	杨晨豪	2017.10.30
校核 Checked	刘士峰	
审核 Reviewed	吴清福	
批准 Approved	杨勇	
比例 Scale		图号 DWG NO.

站内主道路详图

detail of main road of the station

30-B003701S-Z0105-002

版本 REV. A

专业 Speciality	会签 Sign	日期 Date
电气	李艳辉	